How Is GDP per Capita Related to Passenger Car Ownership, and How Strong Is This Relationship Across Countries?

Do richer countries tend to have more passenger cars per person?

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2024 matched UNECE sample

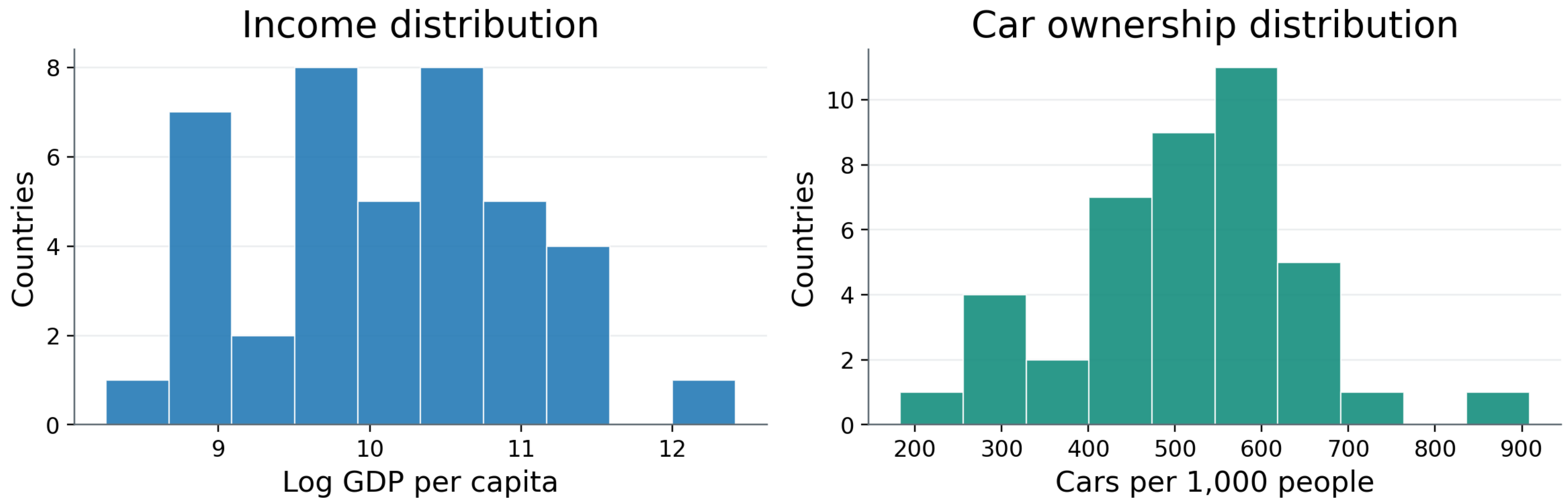
Data and model

VARIABLES

- Y: passenger cars per 1,000 people
- X: log GDP per capita, constant USD
- Sample: 41 matched UNECE countries

Null Hypothesis:
There is no relationship between GDP per capita and passenger car ownership.
$H_0: \beta_1 = 0$
Alternative Hypothesis:
There is a positive relationship between GDP per capita and passenger car ownership.
$H_1: \beta_1 > 0$

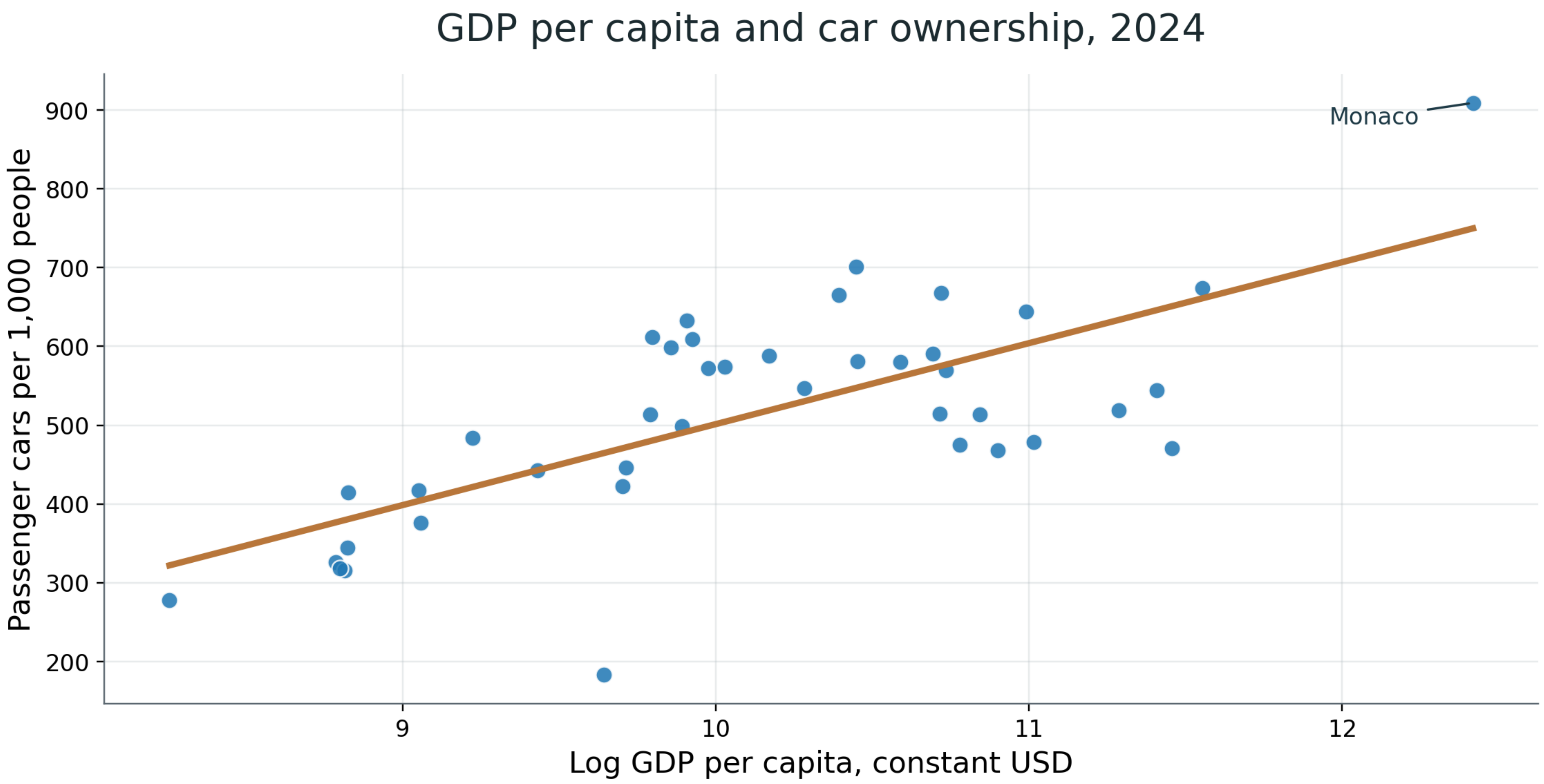
Data snapshot: 41 matched countries



$$\text{Passenger Cars}_i = \beta_0 + \beta_1 \,\text{Log(GDP per Capita}_i) + \varepsilon_i$$

Higher income is associated with higher car ownership

OLS estimate using the 2024 matched sample



Matched UNECE sample, n = 41

102.76

Slope estimate of log GDP

$p < 0.001$

statistically significant

$R^2 = 0.494$

variation explained by X alone

A 10% increase in GDP per capita is associated with about 9.8 more cars per 1000 people.

Conclusion

The evidence supports a clear positive association.

Richer countries in this matched sample tend to have more passenger cars per person.

- Correlational, not causal
- UNECE-based sample, not global
- Monaco strengthens the fitted line
- Future controls: transit, urbanization, fuel prices

Data sources: Our World in Data / World Bank GDP per capita; UNECE passenger car rate table.